HVAC system

create data-Flow diagram for fanSpeedLevel, boost, newSpeed

No loops, in Constructor, one branch for validation of `fanSpeedLevel` testing that creation of super component works correctly, affecting file `SmartDevice.java` rest can be checked by statement coverage

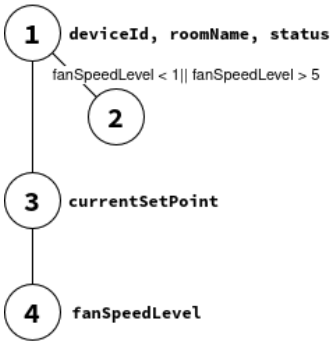

Figure 1: CFG of the Constructor

in adjustFanSpeed, one initialization, three branches, two validating one default outcome as assignment. The Data-Flow for boost is initialization as parameter, usage in statement 1, for newSpeed is initialization in statement one, predicate usage in statement two and 3 and usage in Statement 4, for fanSpeedLevel computational usage in statement 1, assignment in statement 2-4

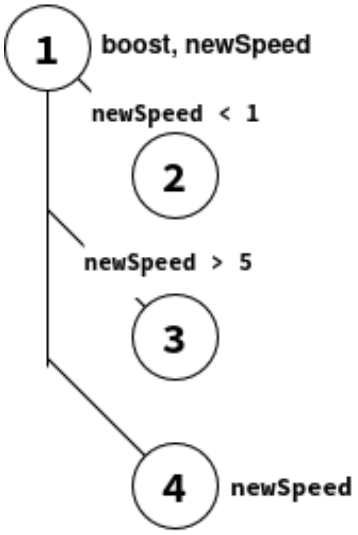

Figure 2: CFG of the adjustFanSpeed method

White-Box-Tests-Constructor

Branch-Test 1
new speed below 1

Branch-Test 2
new speed above 5

Statement-Test 3
new speed in valid range, get currentSetPoint

Statement-Test 4
new speed in valid range, get fanSpeedLevel

White-box-Tests-adjustFanSpeed

Branch-Test 1
boost = −5 Assert getFanSpeedLevel = 1

Branch-Test 2
boost = 5 Assert getFanSpeedLevel = 5

Branch-Test 3
boost = 0 Assert getFanSpeedLevel = previous getFanSpeedLevel

Black-Box-Tests

Robust-Worst-Case BVA
only one input variable with no input boundaries, but fanSpeedLevel has boundaries, leading to 7 data-points,

initial fanSpeedLevel	boost	validity	final fanSpeedLevel
3	−3	`true`	1
3	−2	`true`	1
3	−1	`true`	2
3	0	`true`	3
3	1	`true`	4
3	2	`true`	5
3	3	`true`	5

Weak-Robust ECP
There could be 1 valid class or 5, both versions covered in Robust-Worst-Case BVA